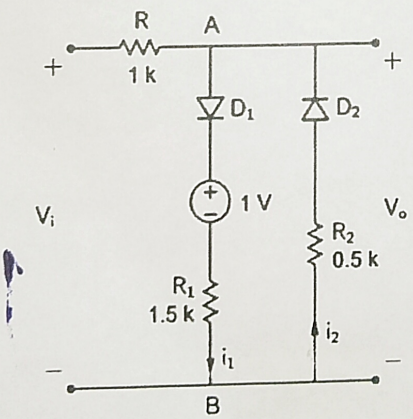
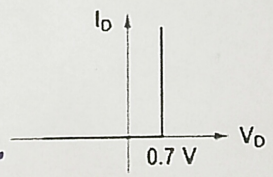
Code No. 3312

OSMANIA UNIVERSITY
FACULTY OF ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (AUTONOMOUS)
B.E. (CSE, AI&ML) III-Semester (MAIN) Examinations DEC-2025/ JAN-2026

BASIC ELECTRONICS

Time : 3 hours **Max. Marks : 60**

Note : *i) Answer **Question No. 1** (Compulsory) and answer any **four questions** from the remaining questions (2- 7). Each question caries 12 marks.*
ii) Answers must be written in same order as they occur in the Question Paper.
*iii) **Missing data, if any may suitably be assumed.***

			Marks	BT	CO
1.	a)	What do you mean by ripple factor ? Give its significance.	2	2	2
	b)	What are the conditions for sustained oscillator ?	2	2	3
	c)	Calculate β when α = 0.98	2	2	3
	d)	Mention the disadvantage of FET compared to BJT.	2	3	4
	e)	Write a short notes on SCR.	2	2	5
	f)	Explain how the FET behaves as VVR.	2	2	4
2.	a)	Explain how zener diode acts as voltage regulator.	6	2	1
	b)	  (a) Plot V_o versus V_i for $-5\ V < V_i < 5\ V$. (b) Plot $V_o(t)$ for a triangular input: −5 V to +5 V, 500 Hz.	6	4	1
3.	a)	Explain the working of Crystal oscillator. Derive the expression for frequency of oscillations of Crystal oscillator.	6	3	3
	b)	Derive the general equation of LC oscillator.	6	3	3
4.	a)	Draw the schematic diagram of summing amplifier. Derive the expression for the output voltage.	6	3	4
	b)	Draw the Full adder circuit using half adder with the logic expression for sum and carryout with truth table.	6	2	4

{P.T.O.}

5.	a)	Sketch and explain the typical shape of drain characteristics of JFET for $V_{GS} = 0V$ with indication of four regions clearly.	6	3
	b)	Explain input and output characteristics of CE configuration.	6	2
6.	a)	Differentiate LED and PHOTO diode.	5	2
	b)	Explain the construction and working of CRO.	7	3
7.	a)	Derive input and output impedance of current shunt feedback.	6	2
	b)	Realize XOR and XNOR using only min NAND gates.	6	3

****